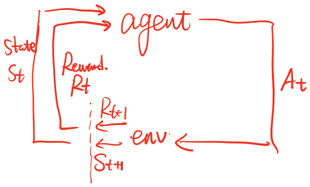


$s \in S$ $a \in A(s)$ $r \in R \in \mathbb{R}$ $p(s', r \mid s, a)$

goal: $\pi(a \mid s)$

$S_0\ A_0\ \boxed{R_1}\ S_1\ A_1\ \boxed{R_2}\ S_2\ A_2 \ldots$

$$G_t = \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k$$

$$\underset{\pi}{\arg\max}\ E_\pi[G_t]$$

$$V_\pi(s) = E_\pi[G_t \mid S_t = s]$$

state value function

$$q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

action value function

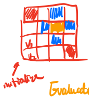
is determinable if we can solve

$$V_*(s) = \max_\pi V_\pi(s)$$

$$q_*(s,a) = \max_\pi q_\pi(s,a)$$

Find optimal π when MDP

Evaluate: compute $V_\pi(s)$ or $q_\pi(s,a)$ given π.



$R_t = 1$ blue $\xrightarrow{\text{update}}$ orange

$\pi(a \mid s) = 0.25$

Reassign value based on Bellman converge

similarity for q

$$\pi_0 \xrightarrow{\text{Evaluate}} V_{\pi_0} \xrightarrow{\text{Improve}} \pi_1 \xrightarrow{E} V_{\pi_1} \xrightarrow{\text{Improve}} \pi_2.$$

policy state values under this policy

policy improvement

$a = \pi(s)$ deterministic

$$\pi_*(s) = \underset{a}{\arg\max}\ q_*(s,a)$$

new policy $\pi'(s) = \underset{a}{\arg\max}\ q_*(s,a)$ $\boxed{\text{greedy}}$ → (2)

policy improvement theorem

keep iterating until no improvement. $\Longleftrightarrow \pi_*$

Assuming MDP: $p(r, s' \mid s, a)$ (agent knows this)

has the access to the environment mechanism.

r \ s	S^{-2}	S^{-1}
0	0.34	0.17
1	0.05	0.23
2	0.17	0.04

	S^1	S^2
0	0.12	0.09
1	.22	0.32
2	0.20	0.05

$\pi(\leftarrow \mid s^0) = 0.4$

$\pi(\rightarrow \mid s^0) = 0.6$

$\gamma = 0.95$

$S_{t+1} \sim p(\cdot \mid s, a)$

$$= E_\pi[V_\pi(S_{t+1}) \mid s^0, \rightarrow]$$

expected over s'

$V_\pi(s')$ By def'n

By Markov

$$E_\pi[G_{t+1} \mid s^0, \rightarrow] = \sum_{s'} p(S_{t+1} = s' \mid s^0, \rightarrow) \cdot E_\pi[G_{t+1} \mid S_{t+1} = s', \cancel{S_t = s^0}, \cancel{A_t = a}]$$

$$E(X \mid Z) = E[E[X \mid Y, Z] \mid Z]$$

Calculate expected future return

By first compute expectations @ each next state. & weight average according to transitional prob

group them into states

$$? \quad E_\pi[G_{t+1} \mid s^0, \rightarrow] = E_\pi[V_\pi(S_{t+1}) \mid s^0, \rightarrow]$$

$$E_\pi[G_t \mid s^0, \rightarrow] = E_\pi[R_{t+1} + \gamma G_{t+1} \mid s^0, \rightarrow]$$

$$= E_\pi[R_{t+1} + \gamma V_\pi(S_{t+1}) \mid s^0, \rightarrow]$$

recursive

$$= \sum_{\substack{r \in \{0,1,2\} \\ s' \in \{s^1, s^2\}}} p(s', r \mid s, \rightarrow)[r + \gamma \boxed{V_\pi(s')}]$$

$$V_\pi(s^0) = E_\pi[G_t \mid S = s_0]$$

law of total prob

$$= \sum_{a \in \{\leftarrow, \rightarrow\}} \underbrace{E_\pi[G_t \mid s^0, a]}_{q_\pi(s^0, a)} \cdot \pi[a \mid s^0]$$

$$= \sum_{a \in \{\leftarrow, \rightarrow\}} E_\pi[G_t \mid S_t = s, A_t = a]\ \pi[a \mid s^0]$$

note $G_t = R_{t+1} + \gamma G_{t+1}$

$$= \sum_{a \in A(s)} \pi[a \mid s^0]\ E_\pi[R_t + \gamma \overset{V_\pi(S_{t+1})}{G_{t+1}} \mid a, s]$$

$$\boxed{= \sum_{a \in A(s)} \pi[a \mid s^0] \sum_{s', r} p(s', r \mid a, s)[r + \gamma V_\pi(s')]}$$

It connect state values.

$V_*(s) = \max_\pi V_\pi(s)$ $q_*(s,a) = \max_\pi q_\pi(s,a)$ Action value

Bellman Optimality $V_*(s^0) = \max_{a \in \{\leftarrow, \rightarrow\}} q_*(s^0, a) = 21.3$

(1) $V_{\pi}(s) = \sum_{a \in A(s)} \pi(a \mid s)\, q_{\pi}(s,a)$ → $V_*(s) = \max_{a \in A(s)} q_*(s,a)$

$$q_{\pi}(s,a) = \sum_{\substack{s' \in S \\ r \in R}} p(s', r \mid s, a)[r + \gamma V_\pi(s')]$$

$$\rightarrow q_*(s,a) = \sum_{\substack{r \in S' \\ r \in R}} p(s', r \mid s, a)[r + \gamma V_*(s')]$$

LHS \ RHS	V_π	q_π
V_π	sub (2) into (1). Bell V	(1)
q_π	(2)	Bell for q_π

Off Policy Much Higher Variance

Behavior Policy: Generates the data $b(a \mid s)$

Target Policy: To be improved/evaluate $\pi(a \mid s)$

On policy	off policy
$b = \pi$	$b \neq \pi$.

$$q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

we'll be estimating

$$E_b[G_t \mid S_t = s, A_t = a]$$

$$q_\pi(s,a) = E_b\left[\underbrace{\frac{p_\pi(G_t)}{p_b(G_t)}} G_t \,\middle|\, S_t = s, A_t = a\right]$$

$$\rho = \prod_{\tau = t+1}^{T-1} \frac{\pi(A_\tau \mid S_\tau)}{b(A_\tau \mid S_\tau)}$$

for Markov

$$P_\pi(\text{trajectory}) = P(S_0 = s_0) \prod_{k=0}^{T-1} \pi(a_k \mid s_k)\, p(s_{k+1}, r_{k+1} \mid s_k, a_k)$$

$$\rho = \frac{P_\pi(\text{trajectory})}{P_b(\text{trajectory})} = \prod_{k=0}^{T-1} \frac{\pi(a_k \mid s_k)}{b(a_k \mid s_k)}$$

everything else env. probability cancel

Monte Carlo Methods

we don't know $p(s', r \mid s, a)$

$S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3$

MC: Averages to Replace the Expected Values

GPI to obtain $\pi \approx \pi_*$ model free

Goal: given π. estimate q_π

$\tilde{S}_t = (S_t, A_t)$

any V estimating $v(\tilde{s}^{new}) \Leftrightarrow q_\pi(s,a)$

all trajectory, all time pt.

$$E_\pi[G_t \mid S_t = s] \approx \frac{1}{C(s)} \sum_{m=1}^{M} \sum_{\tau=0}^{T_m - 1} I[s_\tau^m = s]\, g_\tau^m$$

count how many time state was visited.

$V(s)$

Sample return @ time τ for m^{th} trajectory.

$$g_\tau^m = r_{\tau+1}^m + \gamma r_{\tau+2}^m + \gamma^2 r_{\tau+3}^m + \cdots$$

sum of return in all the data follow state s

Data: M trajectory samples of an MRP

$s_0^m\ r_1^m\ s_1^m\ r_2^m \ \cdots\ S_{T_m}^m$ $m = 1, \cdots, M$

Update Rule to apply after m^{th} sample

$$V(S_t^m) \leftarrow V(S_t^m) + \frac{1}{C(S_t^m)}\left(g_t^m - V(S_t^m)\right)$$

Note: averaging one at a time:

$V_n(s) = \frac{\sum g_i}{n}$ $V_{n-1}(s) = \frac{1}{n-1} \sum_{i=1}^{n-1} g_i$

$$\rightarrow V_n(s) = \frac{V_{n-1}(s) \cdot (n-1) + g_n}{n}$$

$$= (V_{n-1})s - \frac{1}{n}[g_n - V_{n-1}(s)]$$

converting action value → state value

$$q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

define $\tilde{S}_t = (S_t, A_t)$

$$\tilde{V}_\pi(\tilde{s}) = E_\pi[G_t \mid \tilde{S}_t = \tilde{s}]$$

$$= E_\pi[G_t \mid (S_t, A_t) = (s,a)]$$

$$= q(s,a)$$

$\pi(a' \mid s', s, a)$ Markov.

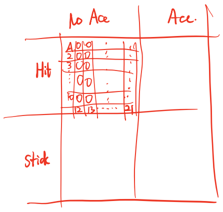
$$\tilde{p}_\pi(s', a', r \mid s, a) = p(s', r \mid s, a)\, \pi(a' \mid s')$$

don't know. but no need.

immediate return

$$Q(s,a) \leftarrow Q(s,a) + \underbrace{\frac{1}{C(s,a)}}_{\alpha}[G_t - Q(s,a)]$$

constant alpha MC.



constant α. MC for estimate $\pi \approx \pi^*$

$\varepsilon > 0$ $\alpha \in [0, 1]$. M total # episodes

Initialize $\pi \leftarrow \varepsilon$-soft policy $Q(s,a)$.

```
for m = 1, 2, ··· M
  Under π. sample: $S_0^m, a_0^m, r_1^m, s_1^m, a_1^m, r_2^m \cdots a_{T_{m-1}}^m, r_{T_m}^m$
    for t = 0, ··· $T_m - 1$:
      $g_t \leftarrow r_{t+1}^m + \gamma r_{t+2}^m + \cdots$
      $Q(s_t^m, a_t^m) = Q(s_t^m, a_t^m) + \alpha(g_t^m - Q(s_t^m, a_t^m))$
                       $= (1-\alpha) Q(s_t^m, a_t^m) + \alpha g_t^m$   new return
    policy improvement:
      $a^* \leftarrow \arg\max_a Q(s_t^m, a)$
```

$$\pi(a \mid s) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|A(s)|} & a = a^* \text{ greedy} \\ \frac{\varepsilon}{|A(s)|} & a \neq a \text{ Any action. choosing uniformly} \end{cases}$$

Check. $(1-\varepsilon) + \frac{\varepsilon}{|A(s)|} \cdot |A(s)| = 1$.

it's a probability

Temporal Difference Learning. Adjust Version of MC

$$\text{Evaluation} \longrightarrow \text{improvement} \longrightarrow E$$

update $V(s)$

Update π based on $\arg\max_a q^*(s,a)$

Adjustment within Here

Adjustment Addresses a disabling feature of MC

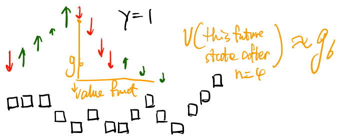
!! An episode must complete before values can be updated

MC delays

terminal state

$$S^+ = \{ s^1, S^2, S^3, s^T \}$$

$$\mathbb{R} = \{ -3, -2, \dots +3 \}$$



$$V(s) \approx E[G_t \mid S_t = s]$$

constant alpha: $V(S_t) \leftarrow V(S_t) + \alpha(g_t - V(S_t))$

$$V(S_t^m) \leftarrow V(S_t^m) + \alpha(\underbrace{g_t^m}_{\text{target}} - V(S_t^m))$$

$$g_{t:t+n}^m = r_{t+1}^m + \gamma r_{t+2}^m + \dots + \gamma^{n-1} r_{t+n}^m + \gamma^n \underbrace{V(S_{t+n}^m)}_{\text{approximate the remaining reward.}}$$

n: lag. • n=1 very eager to make update

$n = \infty$ TD ⟺ MC

bootstrapping

Why does TD perform better than MC

Batch Training

MC.

minimize mean-squared error

just averaging return does not address MRP

Robustness (violation of Markov)

TD's criteria

maximum-likelihood of MRP

$$p(s', r \mid s)$$

$$\hat{p}(s', r \mid s) = \frac{\text{count of } s \to s' r}{\text{count of visits to } s}$$

use $\hat{p}$ to estimat $v(s) = E[G_t \mid S_t = s]$

This is equivalent to 1 step TD

TD's criteria is better for predicting returns

Q-Learning.

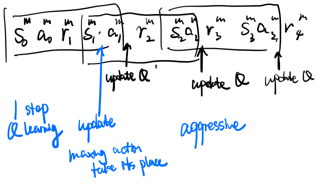
adjustment from 1 step TD

1 step TD $\quad r_{t+1}^m + \gamma Q(S_{t+1}^m, a_{t+1}^n)$

$\downarrow$

Q learning off-policy $\quad r_{t+1}^m + \gamma \max_a Q(S_{t+1}^m, a)$

could be different from policy that generating the data


$S_0^m \; a_0^m \; r_1^m \; S_1^m, a_1^m \; r_2^m \; S_2^m a_2^m \; r_3^m \; S_3^m a_3^m \; r_4^m$
update Q
update Q
update Q
1 step Q learning
update
maxing action take its place
aggressive

Expected Sarsa $\quad r_{t+1}^m + \gamma \max_a Q(S_{t+1}^m, a)$

$\downarrow$

$$r_{t+1}^m + \gamma \sum_a Q(S_{t+1}^m, a)\, \pi(a \mid S_{t+1}^n)$$

next step action values weighted by policy probabilities

on-policy

if $\pi \neq$ generated policy → off policy

Function:

state is not listable

$$v_\pi(s) = E_\pi[G_t \mid S_t = s]$$

$$\approx \hat{v}(s, w) \quad w \in \mathbb{R}^d.$$

Example S is an image

$x_1(S)$: Average of pixel.

$x_2(S)$: sd of pixel.

$$x(S) = \begin{bmatrix} x_1(S) \\ x_2(S) \end{bmatrix}$$

$$\hat{v}(s, w) = x(S)^T w \quad w \in \mathbb{R}^2$$

like $\hat{v}(s,w)$ to match the true value function

$$d \lll |S|$$

Any change to $w \to$ chang $\hat{v}(s,w)$ for many s

Before: only update $V(s)$ for s visited

the best w

distribution over states
↑ # of times in the state
proportional.

$$\overline{VE}(w) = \sum_{s \in S} \mu(s) \left[v_\pi(s) - \hat{v}(s,w)\right]^2$$

SGD: ① states are visited in porportion to μ

② $\hat{v}(s,w)$ is differentiable

③ We observe a surrogate for $v_\pi(S_t)$: $\overset{\text{R.V.}}{\overset{\uparrow}{U_t}}$

$$w \leftarrow w + \alpha \left[\underset{\text{target}}{U_t} - \hat{v}(S_t, w) \right] \nabla \hat{v}(S_t, w)$$

how do we choose ?

If target is Unbiased

$$E[U_t \mid S_t = s] = v_\pi(s)$$

$w \to$ local optimum for $\overline{VE}$

Gradient Monte Carlo

$$U_t = G_t$$

? $U_t = R_t + \gamma \hat{v}(S_{t+1}, w)$

biased $E[U_t \mid S_t = s] \neq v_\pi(s)$

U_t depends on w ???